# PHY 317 – Classical Mechanics – Fall 2026

**Course/contact information**

*Hours:* Mon 1:40–2:55; Wed, Fri 1:20–2:35
*Room:* Sabin-Reed 308
*Credits:* 4
*Text:* Taylor, *Classical Mechanics* (2005)
*Prerequisites:* PHY 210 and PHY 215

*Instructor:* Michael Lerner
*Email:* mglerner@smith.edu
*Office:* McConnell 303
*Website:* Moodle
*Python:* https://jupyterhub.smith.edu

*Office hours* (McConnell 303, shared with PHY 210): Monday 3:00–4:00, Tuesday 1:00–2:00, and Wednesday 4:00–5:00. I also have an open-door policy: stop in whenever my door is open, which is most of the time.

**Course structure**

PHY 317 is Smith's upper-level mechanics course, designed to build on PHY 117 (and a little of PHY 118) and to use the mathematics from PHY 210. We'll follow Taylor: Newtonian mechanics with realistic forces (Ch. 1–4), oscillations (Ch. 5), the calculus of variations and Lagrangian mechanics (Ch. 6–7), central forces and orbits (Ch. 8), non-inertial frames (Ch. 9), coupled oscillators and normal modes (Ch. 11), and chaos (Ch. 12).

The course meets three times a week for 75 minutes, with a lot of group work at the board. There are short pre-class check-ins most days, weekly homework, three in-class exams, and a final exam with redemption problems. The grade breakdown is below. Read the assigned sections *before* class; class time assumes you have.

**Just-in-time review.** The course calendar on Moodle has a "Prerequisites" column that lists, for each new topic, the earlier material it leans on (by Knight chapter for intro physics and by Felder & Felder chapter for math methods). There are two good ways to use it: *just-in-time* – review those concepts before we get to the topic – or *just-after-time* – if a new idea is rough going, use the list to find and shore up the foundation right after class. If you find the prerequisite material itself is the problem, come see me early; closing a gap in week 3 is far easier than closing it in week 10.[1]

**Course learning goals**

By the end of this course, the assiduous student will . . .

- . . . learn, with mathematical rigor, about a variety of features of classical systems, including energy, momentum, oscillations, Lagrangian dynamics, the two body problem, among other subjects.
- . . . use conservation laws (momentum, angular momentum, energy) as *tools*, knowing when each one applies and why;
- . . . explain what chaos is and is not, and recognize it in a simulation;
- . . . use Python to solve equations of motion numerically, to check analytic results, and as a tool for exploration;
- . . . reinforce the assessment toolkit of physics: units, limiting cases, symmetry, and plugging the answer back in.

**Calendar**

The complete day-by-day schedule (topics, reading due, PCCIs, homework due) is posted on Moodle and kept up to date there. In outline:

[1] Much of this syllabus, and much of this course, is taken (with permission) from Will Raven at Smith. Thanks, Will! Smaller bits are taken from Seth Hopper at Earlham. Thanks, Seth!

| Weeks | Topics | Taylor |
|---|---|---|
| 1 | Newton's laws; polar coordinates | Ch. 1 |
| 2 | Projectiles with air resistance; charged particles in magnetic fields | Ch. 2 |
| 3 | Momentum and angular momentum; rockets; center of mass | Ch. 3 |
| 4 | Energy: conservative forces, potential energy, central forces | Ch. 4 |
| 5–6 | Oscillations: damped, driven, resonance, Fourier series | Ch. 5 |
| 7–8 | Calculus of variations; Euler–Lagrange; the brachistochrone | Ch. 6 |
| 8–9 | Lagrangian mechanics: generalized coordinates, constraints, multipliers | Ch. 7 |
| 9–10 | Two-body central-force problem; orbits | Ch. 8 |
| 11 | Non-inertial frames: tides, centrifugal and Coriolis forces | Ch. 9 |
| 12–13 | Coupled oscillators and normal modes; the double pendulum | Ch. 11 |
| 14–15 | Chaos | Ch. 12 |

**Key dates.**

| | |
|---|---|
| Wed Sep 23 | Mountain Day (no class) |
| Mon Oct 12 | No class (autumn recess) |
| Fri Oct 16 | Exam 1 (Ch. 2–4) |
| Mon Nov 16 | Exam 2 (Ch. 5–7) |
| Nov 25, 27 | No class (Thanksgiving) |
| Mon Dec 7 | Exam 3 (Ch. 8, 9, 11) |
| Dec 19–22 | Final exam period (self-scheduled) |

### Grading breakdown

Everything in this course is worth points, and the course total is exactly 1000. Within each category, each assignment is weighted equally. Letter grades are not curved.

| Category | Number | Drop | Each | Total |
|---|---|---|---|---|
| Participation & pre-class check-ins (PCCIs) | 39 | 4 | 3.2 | 112 |
| Weekly homework | 13 | 1 | 24 | 288 |
| Exams (in class) | 3 | 0 | 180 | 540 |
| Final exam, required part | 1 | 0 | 60 | 60 |
| **Course total** | | | | **1000** |

**Grade scale** (percent of 1000; not curved):

| A | A- | B+ | B | B- | C+ | C | C- | D+ | D | D- | E |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 93+ | 90 | 87 | 83 | 80 | 77 | 73 | 70 | 67 | 63 | 60 | $< 60$ |

Each cutoff is the bottom of that grade (a B- is 80–82.9, and so on).

### Pre-class check-ins and attendance

Most class days there will be a short PCCI due at the start of class, on paper: usually one Taylor problem, chosen so that working it primes you for that day's material. **You will receive full credit for a good-faith effort that includes a clear description of anything you are stuck on.** An incorrect or incomplete answer *without* such an explanation won't receive full credit. For example, if the question were to differentiate $f(g(x))$: "I don't know" earns nothing; the correct answer earns full credit; and so

does "I think it's $f'(g(x))$, except I know the chain rule says to do something different because it's $f(g(x))$ and not just $x$ – I'm not clear on what, though." A PCCI should take you less than 15 minutes; if one doesn't, please tell me.

Turning in the day's PCCI is also how attendance is recorded, and your four lowest participation days are dropped, so occasional absences and off days cost you nothing. This is a small class, and it depends on group work at the board; your classmates are a resource for you and vice versa, and that in-class work is one of the main benefits of a school like Smith. If you have an accommodation, please come talk to me so that we can make the class work for both of us. *In general*, I will expect you to miss no more than four classes, and missing more than four may adversely affect your grade.

- if you miss more than **4 classes**, the highest grade you can earn is a **B**.
- if you miss more than **8 classes**, the highest grade you can earn is a **C**.
- if you miss more than **12 classes**, the highest grade you can earn is a **D**.
- if you miss more than **16 classes**, you will fail the course.

Please prioritize consistent attendance to support your learning and success in this course.

Additional absences due to emergencies, illnesses, or other extenuating circumstances can be approved for this class through conversations with your class Dean.

HOMEWORK

We learn physics by doing problems, and upper-level mechanics problems are long: expect a set of five or six Taylor problems each week, most of them substantial. Homework is posted on Moodle and due there **Fridays by 10:00 PM** (usually) as a single PDF of your written work (photos of handwritten pages are fine; combine them into one PDF). I move a few due dates around due to exams and breaks, but Moodle's due dates count as authoritative. Most weeks one problem is **computational**, done in Python on jupyterhub.smith.edu (see *Python* below); include your code and its plots in the PDF.

You may (and should) work with others on the homework, but what you submit must represent your own understanding, written up by you. Your lowest homework score is dropped.

**Every problem ends with an assessment.** After the answer, write two or three lines checking it: a limiting case ($m \to 0$, $t \to \infty$, the angle going to zero), a units check, or a comparison with a result you already trust. Some problems on the sets say "include an assessment" or "this is your assessment"; the habit is expected on all of them, and it is worth points in the rubric.

**Grading methodology for homework:**

$\checkmark^+$ = 100%: The problem, including units and assessment when appropriate, is completely correct.

$\checkmark$ = 95%: There was a small problem.

$\checkmark^-$ = 50%: There were multiple small problems or a big problem. Submitting a completely correct solution along with a few sentences reflecting on the problem increases your grade up to an 85%.

X = 25%: There was a lot of confusion on the topic. Submitting a completely correct solution along with an in-depth reflection on the problem increases your grade up to a 75%.

0: You skipped the problem. **Don't do this!** You get 25% just for writing *something* down!

Your reflection should address why you got the problem wrong, your initial misconceptions, and, most importantly, what you have learned to correct those misconceptions. Research in pedagogy highlights reflection as one of the most effective learning tools. Resubmissions and reflections are due within **10 days after the homework is returned** and go in the same PDF as the next homework set, clearly labeled. Reflections that lack sufficient depth or fail to demonstrate meaningful engagement with the problem will result in fewer points.

LATE POLICY

It is extremely hard to catch up on late work (usually things are late because you are busy, and people don't tend to get less busy!). That is why every category has built-in drops: four participation days, one full homework set. My strong suggestion, when life happens, is to use the drops and prioritize getting your schedule back under control.

Unless approved beforehand *late homework will receive at most half credit.* Homework greater than one week late will not be accepted.

EXAMS

Three exams, in class, taking the full period, on the dates above. Each exam has **three problems, one per chapter covered**; a student with command of the material should finish each in about 25 minutes. Computational tools (Mathematica/Python) may be permitted for specific problems, which will be indicated on the exam itself.

**Grading and redos.** Each exam problem gets one mark from the homework scale above ($\checkmark^+$, $\checkmark$, $\checkmark^-$, X, 0), scaled to the problem's 60 points. For any problem marked $\checkmark^-$ or X, you may submit a completely correct solution with a reflection, exactly as for homework, and the same ceilings apply: 85% for a $\checkmark^-$, 75% for an X. Exam redos are due **one week after I hand the exam back**, on Moodle as a single PDF; the due date goes on the Moodle calendar when I return the exam. You may use the book, your notes, your classmates, and office hours to understand where you went wrong, but write the redo up on your own, without AI. Redos and the final's redemption problems are independent: whichever score is higher stands.

**The final exam**, during the December 19–22 self-scheduled exam period, has two parts:

1. **Required** (60 points): one problem on Chapter 12 (chaos) is mandatory.
2. **Optional grade-replacement problems**: nine more problems, one for each chapter that appeared on the three exams (Chapters 2–9 and 11). You may redo **up to four of them in total**. For each one you solve, the new score *replaces* your score on the corresponding exam problem if and only if it is higher. There is no penalty for an attempt; this can only improve your grade.

Do the required problem first; then pick up to four redemption problems that you want and have time for.

**Partial credit, and how to get it.** Just like with homework problems, make sure to write *something* down. The more of your thought process you write down, the more credit I can give you. Assess things. If your assessment says you got something wrong, but you don't know what you got wrong *say that*!

PYTHON (AND MATHEMATICA)

Computation is part of doing mechanics: the equations of motion for quadratic drag, the driven pendulum, and the double pendulum have no closed-form solutions, and numerical solutions are how physicists actually see those systems. We will use Python (NumPy, SciPy, SymPy, Matplotlib) in Jupyter notebooks on https://jupyterhub.smith.edu, the same setup as PHY 210. I'll post starter notebooks for the computational homework problems and for the in-class simulations. Previous offerings used Mathematica, and Smith has a license; if you already know it and prefer it, you may use it for any computational problem.

Two expectations:

1. **Simplification first.** For homework and exams, anything that can be done with concepts – an integral that vanishes by symmetry or because the integrand is odd, a limit you can read off – should be done on paper first. Computational tools can be used to confirm these results, but should not replace your understanding of the core principles.

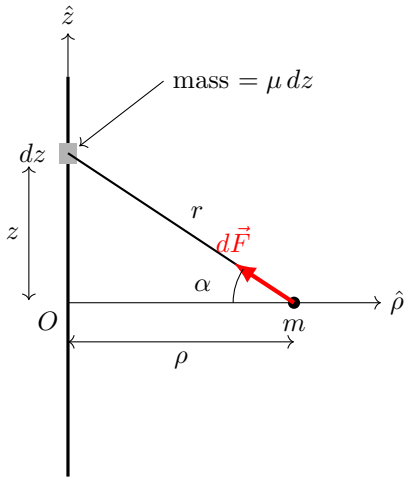
2. **Exploration and assessment.** Use computational tools to push past the algebra: vary a parameter, plot something, check a limiting case numerically. Say in your write-up what you predicted *before* running it and what you concluded *after*.

**Example homework problem using Python.** *Consider an infinitely long, uniform rod of mass $\mu$ per unit length that lies along the $z$-axis. Calculate the gravitational force on a point mass $m$ at a distance $\rho$ from the $z$-axis.*

*Solution.* First, I want to consider the force on $m$ due to a short segment $dz$ of the rod at a height $z$ above $m$. The force has magnitude

$$dF = \frac{Gm\mu\,dz}{r^2}, \qquad r = \sqrt{z^2+\rho^2},$$

in the direction shown in this picture:




To find the total force, I need to integrate from $z = -\infty$ to $+\infty$. When we do this, the $z$-components $dF_z$ will cancel due to symmetry: the segment at $-z$ pulls up exactly as hard as the one at $+z$ pulls down. In addition, the component into the page must also be zero due to cylindrical symmetry. So I just need to find the radial component,

$$dF_\rho = -\,dF\cos\alpha = -\frac{Gm\mu\,dz}{r^2}\,\frac{\rho}{r} = -\frac{Gm\mu\,\rho\,dz}{(z^2+\rho^2)^{3/2}},$$

with the minus sign because it points toward the rod (the $-\hat{\rho}$ direction). Since we are integrating over $z$, I have expressed $r$ in terms of $z$; note that $\rho$ is independent of $z$. Then

$$F_\rho = -Gm\mu\rho\int_{-\infty}^{\infty}\frac{dz}{(z^2+\rho^2)^{3/2}} = -\frac{2Gm\mu}{\rho}.$$

I used Python for the last step – SymPy for the integral itself – and then, just for fun, I had it confirm that the $z$-component really is zero. It must be, since $dF_z = dF\sin\alpha = Gm\mu\,z\,dz/r^3$ is odd in $z$:

$$F_z = Gm\mu\int_{-\infty}^{\infty}\frac{z\,dz}{(z^2+\rho^2)^{3/2}} = 0.$$

```
import sympy as sp
G, m, mu, rho = sp.symbols("G m mu rho", positive=True)
z = sp.symbols("z", real=True)
r = sp.sqrt(z**2 + rho**2)
F_rho = sp.integrate(-G*m*mu*rho / r**3, (z, -sp.oo, sp.oo))
F_z   = sp.integrate( G*m*mu*z   / r**3, (z, -sp.oo, sp.oo))
print(F_rho, F_z)     # prints:  -2*G*m*mu/rho   0
```

The final answer is then

$$\vec{F} = -\frac{2Gm\mu}{\rho}\,\hat{\rho}.$$

*Assessment:* the units work ($G\mu m/\rho$ has the units of $GMm/r^2$ with $M = \mu\rho$), and the $1/\rho$ falloff (rather than $1/\rho^2$) is what an infinite line source should give – the same thing happens for the electric field of an infinite line charge in PHY 118. While I plugged the second integral into SymPy, it must be zero since the integrand is odd; the computer is confirming the symmetry argument, not replacing it.

### Inclusion and equity

I can hardly overstate how much inclusion, equity and diversity considerations have shaped my entire thinking around teaching and practicing physics. It affects everything from the way that physics is framed historically to the way that it is typically taught in your classrooms. My main goal is to create an environment that tells all of us that we can do physics; that invites people into physics who have never been invited, or who have been explicitly excluded.

Physics cannot be divorced from the world in which we study it. I expect that we will recognize that in this class, and I expect that we will create an inclusive and welcoming environment. More specifically, I expect that we will strive *not* to create an environment that excludes people. This is hard work. We will get things wrong. You will get things wrong. *I* will get things wrong. Especially when I get things wrong, I encourage you to tell me, using any of the channels under *Sensitive issues* below.

### Use of artificial intelligence

I find AI to be a pretty cool and powerful tool, and it offers incredible potential to enhance your learning experience in this course. However, like any powerful tool, it needs to be used wisely. Used improperly, it can genuinely hinder your development as a physicist.

You are welcome to use AI in this class, provided you always adhere to the Academic Integrity Statement (below). Crucially, **using AI to directly solve homework or exam problems is a clear violation of the Academic Integrity Statement and will be treated as such.** Your goal here is to learn, to grapple with challenging concepts, and to develop your own problem-solving abilities. AI can be a fantastic accelerator for that learning, but it cannot replace the essential process of your own intellectual struggle and discovery.

Be aware that AI models, while impressive, are not infallible. They can (and often do) make subtle errors, "hallucinate" incorrect information, or provide solutions that, while technically correct, don't foster the deeper understanding you need. Always approach AI-generated content with a critical eye, just as you would any other resource. Your job is to verify, question, and ultimately understand the physics.

I strongly encourage you to experiment with AI as a study partner. It can be a powerful resource for clarifying concepts, debugging code, or practicing your critical thinking. Think of it as a sophisticated tutor available 24/7. Below are some recommended ways to start using AI effectively. (Two additions: PCCIs are graded on good-faith effort, so the only thing being graded is a record of *your* thinking, and having AI polish that helps your grade not at all; and early in the semester we will spend part of a class deciding

*together* how AI should and shouldn't fit into homework in this course, computational work especially, and post what we decide on Moodle next to this policy.)

**Recommended AI prompts for learning.** When using AI, it's helpful to give it context. Always start by telling the AI who you are and what you're studying.

*Initial context-setting prompt (always include this first):* "I am a physics major taking upper-level classical mechanics at Smith College. I am currently studying [specific topic, e.g., Lagrangian mechanics, coupled oscillators]. Please tailor your responses to this context, assuming I have a weak/medium/strong foundation in calculus, differential equations, and introductory physics."

1. **For conceptual understanding and definitions (ask the AI to ask *you* questions).** Use this when you want to solidify your grasp of a concept or definition. Instead of just getting an answer, prompt the AI to help you critically evaluate your own understanding. "I'm trying to deepen my understanding of [specific topic, e.g., time-dependent potential energy] in classical mechanics. Here's my current understanding/definition: [your definition]. Instead of just giving me a refined definition, can you ask me a series of probing questions that would help me critically evaluate and improve my understanding, focusing on the fundamental implications and common misconceptions related to [topic] and its role in classical mechanics?"

2. **For programming help (focus on debugging skills, not just fixes).** When your code isn't working, use AI to learn how to debug systematically. "I'm working on a [Python, Mathematica] program to model [what it does, e.g., the motion of a 2D projectile subject to quadratic drag], and I'm encountering issues. My goal is not just to fix the code, but to understand the debugging process. Here is my program: [code]. Can you help me identify potential errors by explaining your diagnostic process rather than just providing the corrected code? Focus on common pitfalls in physics simulations and how to systematically approach debugging them. If there are multiple errors, address them one at a time."

3. **For refining reflections and critical thinking.** This is excellent for analyzing why you made mistakes and how to avoid them in the future. "I'm preparing a reflection for a classical mechanics problem where I struggled. The purpose of this reflection is to critically analyze my mistakes and articulate what I've learned to prevent similar errors in the future, thereby strengthening my problem-solving and critical thinking skills as a physicist. Here's the problem: [problem]. My primary point of confusion or error was: [specific confusion/error]. Can you act as a mentor and ask me a series of thought-provoking questions that will guide me in drafting a comprehensive and insightful reflection? Focus on probing the *conceptual* rather than just computational errors, and push me to think about the underlying physics principles and problem-solving strategies."

4. **For conversational exploration of confusing topics.** When a topic just isn't clicking, use AI as a dialogue partner to explore it in depth. "I want to have a detailed, high-level conversation about [confusing topic, e.g., time-dependent potentials] in classical mechanics. My goal is to develop a robust conceptual and mathematical understanding. Can you start by explaining a key fundamental aspect or a common misconception, and then we can explore it through a dialogue? Feel free to challenge my assumptions or prompt me with scenarios."

5. **"What if?" scenarios (predict and explore variations).** Challenge yourself by exploring how changes to a problem affect the outcome. "I've just solved a classical mechanics problem involving [e.g., a simple pendulum with damping]. To deepen my understanding, can you propose a 'what if' scenario that slightly

modifies the problem parameters or introduces a new element, and then ask me to predict the qualitative (and perhaps quantitative) impact? We can then discuss the implications and verify my reasoning. The goal is to explore the boundaries of my understanding of [relevant physical principle]."

**Communication**

My preferred mode of communication is email. I will check my email multiple times per day during the work week. If you have sent me an email and have not heard back from me within 24 hours (excluding weekends), you should email me again or drop by my office to ask your question in person. I cannot guarantee that I will regularly check my email on evenings or on weekends.

I expect that you will check your Smith email at least once per day: 24 hours after I email the class, I expect everyone has read it (excluding weekends, of course). Emailed assignments, due dates, and policy changes carry the same weight as this syllabus, and the same goes for Moodle.

**Workload.** This course should take about 12 hours of your time per week: about 4 in class, and about 8 outside it on reading, PCCIs, homework problems, and review. **If you find yourself spending significantly more (or less) than this, please let me know ASAP! That's an instructor-error that I can easily fix.**

**Sensitive issues**

What if you have something you want to talk about, but don't feel comfortable doing so in a public (or non-anonymous) space? This often comes up with issues of racism, sexism, and other kinds of discrimination. It can come up with issues that relate to my behavior, the behavior of your classmates, or the world in general. Please feel *encouraged* to use either the anonymous feedback form linked on Moodle, or to leave a written note under my door (if you want to write a note, you may want to type it up and print it out so that your handwriting is further anonymized). Of course, if you feel comfortable and safe doing so, you are also encouraged to talk publicly or non-anonymously about these things! If you are not comfortable with any of these options, the *Getting help* section below lists offices that can help, including confidential ones.

**Recording**

My plan is to record from my iPad to both PDF and videos, posted to Moodle. We'll see how technology works. I do this because several students have used it as a post-class resource in the past. Sometimes a derivation is worth reviewing, sometimes you were having a spacey day, etc. But it's worth knowing that student experience shows that it's a very poor substitute for attendance, and just watching the recordings won't count towards your 112 attendance/participation points. Sometimes the mic picks up conversations in the room. If that's a problem, I can look into muting it. Let me know.

**Academic integrity**

Real learning happens when students conduct themselves with academic integrity, and our goal here is to provide you with the opportunity for (and expectation of) that learning. Read the Academic Integrity Statement. That statement is a gift to the community, giving you all sorts of privileges (self-scheduled finals, the expectation that I treat you like academic adults, etc.). And it comes with responsibilities. All submitted work of any kind must be the original work of the student, who must cite all the sources used in its preparation – including, for example, an AI model.

In this class specifically: you are encouraged to work together on the homework and to study together for exams, but PCCIs should be attempted individually first, everything you turn in must be your own work, and exams and the final must be done individually. If you cannot explain why you have written what is on your paper, you have crossed from collaborating to copying.

**Academic accommodations**

If you need classroom accommodations, please contact the Accessibility Resource Center and file things through the Accommodate system. I promise to be flexible in meeting any documented need for accommodations in a way that works well for you. If you would like some accommodation but do not have an ARC reference, let me know and we can discuss your concerns: regardless, I want to provide you the support you need to succeed in this course, though I will do so in a way that does not diminish the accommodations of others. Remember to contact me well in advance of when you need the accommodations in place, so that we can prepare.

**Course feedback**

There will be an anonymous mid-semester feedback form on Moodle around fall break, and during the last week of classes approximately 15 minutes at the beginning of one class will be set aside for you to complete the official course feedback questionnaire. Please don't wait until then if something isn't working.

**Getting help**

**People and places.** There are many resources at Smith to help you succeed in this course and in your college career:

- Your instructor! Think of my office hours as **free help sessions**: if you are going to spend a couple of hours a week on homework anyway, why not do some of it where the person who writes your exams can help? That's my job, so don't think it's bothering me to have you reach out with questions or concerns.
- The Spinelli Center – open hours, one-on-one appointments, and math review workshops.
- Writing help, learning-specialist services, and workshops on time management and managing stress, all from the Jacobson Center.
- Crisis resources, counseling, and wellness services at the Schacht Center.
- Resources for gender identity and expression, and the Office for Equity and Inclusion in general.
- Where to report sexual misconduct and other discrimination. Please note that your instructors are Responsible Reporters.